Will Cheaper Machine Intelligence Raise Employment? Demand Expansion, Substitution, and the Early Evidence

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Working paper, October 2026

Abstract

Does cheaper machine intelligence raise or lower employment? In the theory of derived demand, a fall in the price of an input that can replace labor raises employment if and only if the price elasticity of product demand exceeds the elasticity of substitution between that input and labor—the condition Hicks identified when qualifying Marshall's third law. Applied to artificial intelligence, the condition yields three conclusions. The boom in AI use as prices have fallen signals that AI substitutes readily for labor, which raises rather than lowers the bar for job creation. The absence of aggregate employment effects since 2022 is what the condition predicts while AI's share of costs remains small, whichever view is right. And the finding that novices gain most from AI assistance is consistent with falling entry-level hiring, because AI used as a copilot and AI used as an agent are governed by different elasticities. I extend the condition to allow for new tasks, which give employment a second route to growth that the pessimistic view also contests, classify twelve historical episodes of automation, show how both elasticities can be measured, and state predictions, fixed in advance, for 2028 and 2031.

Since late 2022, large language models and related artificial intelligence (AI) systems have sharply reduced the cost of performing many cognitive tasks. In the business market for model inference, prices fell by roughly three orders of magnitude in about two years (Demirer, Fradkin, and Tadelis 2026). Whether this will reduce employment in the occupations that perform those tasks is among the most consequential open questions in economics, and public debate has framed it as a contest of narratives. On one account, cheaper intelligence will be used far more, as cheaper calculation was after the electronic spreadsheet, and demand for the workers who produce cognitive services will expand with it. On the other, this technology differs from its predecessors because it can perform whole jobs, so expanded demand will be met with more machine services rather than more workers. Geoffrey Hinton, who has made the second argument publicly (Hinton 2025b; Ma 2025), put it to me in six words: "The new hires will be AI" (personal communication, 2026).

The question is old, and so is the apparatus for answering it. Ricardo added a chapter "On Machinery" to the third edition of his *Principles* in 1821 to concede that mechanization could harm workers (Hollander 2019), and each later wave of automation has revived the concern (Mokyr, Vickers, and Ziebarth 2015). For a single market, the theory of derived demand gives a precise answer: when the price of an input that can replace labor falls, employment rises if and only if the price elasticity of product demand, ε, exceeds the elasticity of substitution between that input and labor, σ. This is the inequality that Hicks (1932) showed governs Marshall's (1890) third law

of derived demand, and it is a textbook result in the theory of labor demand (Hamermesh 1993). The two narratives about AI are therefore claims about two elasticities—and, over longer horizons, about whether new tasks for people appear as fast as AI takes over existing ones. Stated that way, the dispute becomes partly testable, and the evidence can be sorted by what it does and does not identify.

Applied to AI, the condition yields three conclusions that run against common readings of the evidence. First, the rapid growth in the use of AI as its price has fallen—often described as a Jevons paradox, after Jevons's (1865) observation that more efficient steam engines raised Britain's consumption of coal—is not evidence that AI will create jobs. Total use of an input rises with its efficiency if and only if it substitutes readily for the inputs it works with; while AI's cost share is small and labor is the main other input, that means roughly $\sigma > 1$. To the extent that the boom in AI use reflects falling prices, it therefore raises the bar for the optimistic view: employment grows only if demand for the product is more elastic still. The familiar argument that elastic demand turns productivity gains into jobs is valid for AI that makes workers more productive, not for AI that replaces them.

Second, the near-zero aggregate employment effects reported since 2022 are what the theory predicts whichever view is right. In the framework, the employment effect of cheaper AI services is scaled by AI's share of production costs, and the effect of AI copilots by the productivity gains they deliver. By both measures, AI's role in production remains small: by the second quarter of 2026, generative AI was saving about 2 percent of all US work hours (Federal Reserve Bank of St. Louis 2026). The null results rule out rapid disruption, but they carry little information about the sign of $\varepsilon - \sigma$. That AI's role is still so small after a thousandfold fall in list prices is itself informative: unless AI substitutes only weakly for labor, the effective price of AI services—the cost per task completed, including the cost of integrating AI into production and checking its work—has probably fallen far less than the price of inference.

Third, the framework offers a resolution of a puzzle in the evidence. Field experiments find that less experienced or less skilled workers gain the most from AI assistance, yet US payroll data show the clearest declines in the employment of young workers in exposed occupations, mostly through reduced hiring. The experiments measure AI as a copilot that raises the productivity of people already employed; hiring responds to AI as an agent that performs tasks at a falling price. The two channels are governed by different conditions, and the capability that helps a novice most—access to codified expertise—can also be the capability that lets a firm do without one.

The article builds on earlier contributions to the *Journal of Economic Perspectives*. Autor (2015) argued that automation's effect on the demand for workers depends on how strongly automated and human tasks complement each other, on the elasticity of labor supply, and on the price and income elasticities of demand for the output. Acemoglu and Restrepo (2019) decomposed technology's effects on labor demand into displacement, productivity, and reinstatement effects, and found that since the late 1980s displacement has accelerated while reinstatement has slowed. Agrawal, Gans, and Goldfarb (2019) showed that the labor-market effect of AI that automates prediction is ambiguous, because cheaper prediction substitutes for workers in prediction tasks but comple-

ments human judgment and can make it profitable to automate whole decisions. In the journal's recent symposium on artificial intelligence, Jones (2026) argued that weak links among complementary tasks will slow AI's effect on output, and Demirer, Fradkin, and Tadelis (2026) described the market in which firms buy AI; Jones treats labor markets only briefly, and Demirer, Fradkin, and Tadelis leave them outside their scope. Beyond the three conclusions above, this article extends the condition to allow new tasks—a second route to employment growth even when $\sigma > \varepsilon$—classifies twelve historical episodes of automation by the scope of the automated tasks and the response of demand, shows how ε and σ for AI can be measured with data that exist or could be collected, and states predictions with numerical thresholds, fixed before the data exist, in the spirit of the pre-analysis plans discussed in the same journal (Olken 2015; Coffman and Niederle 2015).

The evidence through mid-2026 supports a guarded reading. Demand for cognitive services is plausibly elastic where current prices exclude much of the potential demand, as in civil legal services for low-income households, but there is no evidence that it is elastic across cognitive work generally, and the historical record suggests that demand elasticities fall as markets mature (Bessen 2019). In the episodes examined here, automation raised incumbent employment when it automated part of a job in a market with expanding demand, and lowered it when it replaced the whole job, whatever happened to demand. The decisive questions—whether σ rises past ε as AI capabilities broaden, and whether new tasks for people appear faster than AI absorbs them—cannot yet be answered with economic data, but the indicators that should reveal the direction of change within about two years can be specified now.

Derived Demand: An Old Answer to a New Question

The Condition

Consider a competitive industry that produces output with constant returns to scale from two inputs: labor and AI services, meaning model inference, compute, and software agents purchased as inputs. Each input has a productivity term that converts physical units into effective units, a_L for labor and a_M for AI services. The industry takes the wage, w, and the price of AI services, p_M, as given and prices output at unit cost, so cost reductions pass through fully to the output price. Product demand has a constant price elasticity, ε, measured as a positive number. Let s_M denote AI's share of production cost and σ the elasticity of substitution between effective labor and effective AI services. The percentage change in employment, L, in response to a 1 percent change in the effective price of AI services, p_M/a_M, is

$$\frac{\partial \ln L}{\partial \ln(p_M/a_M)} = s_M\,(\sigma - \varepsilon).$$

When AI services become cheaper, firms substitute them for labor at a rate governed by σ. The cost saving also lowers the output price, which raises the quantity demanded at a rate governed by ε. Employment rises if and only if the second effect is larger, that is, if $\varepsilon > \sigma$. Both effects are scaled by AI's cost share, so while AI accounts for a small share of costs, a modest fall in its price has a small

effect on employment in either direction. The result is standard (Hamermesh 1986, 1993). For the large price declines of recent years, its counterpart is that the cumulative change in employment is of the order of the change in AI's cost share unless σ is close to one or ε differs from σ by much more than σ differs from one; online Appendix A derives both results in this notation.

Copilots and Agents

AI can also enter production as a tool that raises the productivity of the workers who use it, a_L, without being a separately priced input. In the same setting, the employment response to such labor-augmenting technical change is

$$\frac{\partial \ln L}{\partial \ln a_L} = s_L\,(\varepsilon - 1) + s_M\,(\sigma - 1),$$

where $s_L = 1 - s_M$ is labor's cost share. When AI is not a separate input ($s_M = 0$), the expression reduces to $\varepsilon - 1$: higher productivity raises employment if and only if product demand is elastic. This is the logic behind most public claims that cheaper AI will create jobs.

I refer to the two ways AI enters production as the copilot channel and the agent channel. A copilot makes the worker who uses it more productive; an agent performs a task itself, at a price. The demand-expansion view assumes, usually implicitly, that AI operates mainly as a copilot, or as an agent with $\varepsilon > \sigma$. The substitution view holds that AI increasingly operates as an agent with $\sigma > \varepsilon$, and that σ rises as AI performs a growing share of each occupation's tasks. Once AI is a separately priced input, elastic demand is neither necessary nor sufficient for employment to rise. It is not sufficient because σ may exceed ε, and it is not necessary because $\varepsilon > \sigma$ can hold with $\varepsilon < 1$ when AI and labor are poor substitutes.

The agent channel has a limiting case. If AI services and labor are perfect substitutes in a task, firms use labor in that task only when its effective cost is no higher than that of AI, so the wage for the task cannot exceed $(a_L/a_M)\cdot p_M$. At this corner the local formulas do not apply; instead, the cost of AI caps what labor can earn in the task, and the cap falls with the price of AI. Restrepo (2025) obtains the economy-wide counterpart: once compute can perform all the work that is essential for growth, wages converge to the cost of the compute needed to reproduce human work.

When the List of Tasks Changes

The condition takes the set of tasks as given, but technology also changes which tasks exist and who performs them. This opens a second route by which employment can rise or fall. Write the constant-elasticity technology in the form used by Acemoglu and Restrepo (2019), and let Γ be labor's weight in production—what they call the task content of production for labor. Holding the wage and labor productivity fixed, employment then changes by

$$\Delta \ln L \;=\; \underbrace{(\sigma - \varepsilon)\,\Delta \ln p}_{\text{existing tasks}} \;+\; \underbrace{\Delta \ln \Gamma}_{\text{task content}} \;,$$

where Δln p is the change in unit cost from any source (online Appendix A). The first term is the condition above: the race between substitution and demand expansion within a given set of tasks. The second records changes in the tasks assigned to labor. It falls when AI takes over tasks that people performed, which Acemoglu and Restrepo call displacement, and it rises when new tasks appear in which people have the advantage, which they call reinstatement. Γ is not a third elasticity; it summarizes how innovation changes the structure of production.

The extension makes three points precise. First, employment can rise even when $\sigma > \varepsilon$, if new tasks for people raise Γ fast enough, and it can fall despite elastic demand if AI takes over tasks faster than new ones appear. Second, new products are not the same as new tasks. A new product made with existing tasks raises the demand for those tasks, an effect that runs through ε; only new tasks in which people have a role raise Γ. Third, the substitution view challenges both routes. "The new hires will be AI" asserts both that AI will meet expanded demand, a high σ, and that AI will perform the new tasks as well, so that reinstatement stays small. The second claim is what would set AI apart from earlier technologies, which created work they could not do themselves: photography needed photographers, and computers needed programmers.

What a Boom in AI Use Reveals

The growth in AI use as prices have fallen is often cited as evidence for the demand-expansion view, by analogy with Jevons's observation about coal. The share of employed US adults who used generative AI for work in the previous week rose from 28 percent in the third quarter of 2024 to 39 percent in the second quarter of 2026 (Federal Reserve Bank of St. Louis 2026). The framework shows that a Jevons effect in AI use speaks to σ, not to ε. The own-price elasticity of demand for AI services is $-(s_L{\cdot}\sigma + s_M{\cdot}\varepsilon)$, and total use of AI services rises with their efficiency, a_M—equivalently, spending on them rises as their price falls—if and only if $s_L{\cdot}\sigma + s_M{\cdot}\varepsilon > 1$ (online Appendix A). With AI's cost share small, this requires approximately $\sigma > 1$. When AI works with several other inputs, the condition concerns its substitutability for all of them together, which can reflect substitution for software or purchased services as well as for labor. And adoption shares like those above rise with any downward-sloping demand; the informative statistic is spending on AI services as their effective price falls.

Spending on AI services that rises as their effective price falls would therefore be evidence that AI substitutes readily for labor. It would not settle the employment question, because employment can still rise if $\varepsilon > \sigma$, but it would raise the bar: with $\sigma > 1$, that requires $\varepsilon > 1$ as well. Jevons's paradox concerned an input whose efficiency rose. Its analogue for workers is the copilot channel, where the condition for employment to rise is $\varepsilon > 1$, so the analogy supports the demand-expansion view only to the extent that AI raises the productivity of workers rather than performing their tasks. Growth in AI use also reflects diffusion and an expanding set of tasks that AI can perform at all,

and both weaken the inference from use to σ: diffusion is not a response to price, and an expanding set of tasks that AI performs is a fall in Γ rather than a movement along the demand curve for AI services. Growth of the second kind is nonetheless evidence of displacement, because in task-based models the movement of tasks from labor to capital is the main channel through which automation displaces workers (Acemoglu and Restrepo 2018).

What the Condition Leaves Out

The formula describes one market in partial equilibrium, and four qualifications matter for interpreting the evidence.

First, the two elasticities must refer to the same market. A single firm's demand elasticity includes business taken from rivals and so exceeds the market elasticity. French firms that adopted robots expanded employment while employment in their industries fell (Acemoglu, Lelarge, and Restrepo 2020; see also Koch, Manuylov, and Smolka 2021), so evidence that AI-adopting firms are hiring is weak evidence that $\varepsilon > \sigma$ for their industries.

Second, σ must be interpreted with care. In task-based models where each task is performed by either capital or labor, the elasticity of substitution between the two at a given assignment of tasks equals the elasticity of substitution across tasks—the parameter that Jones (2026) argues is below one. Displacement in those models comes instead from the extensive margin, as capital takes over tasks that labor used to perform (Acemoglu and Restrepo 2018, 2019). In the reduced form used here, two cases differ. When cheaper AI takes over more of the tasks it can already perform, the effective σ exceeds the cross-task elasticity. When the frontier of capabilities advances, the reduced form registers a fall in labor's task content, Γ. The constant-σ formula is best read as describing the response of employment to cheaper AI at a given frontier; how the frontier moves is taken up later.

Third, aggregate effects differ from industry effects. With a fixed labor supply and flexible wages, a fall in labor demand in one industry shows up mainly as lower relative wages and the reallocation of workers. Caselli and Manning (2019) show that under constant returns, perfect competition, and a constant interest rate, a new technology cannot reduce the average real wage in the long run unless it raises the price of investment goods relative to consumption goods, although particular groups of workers, even a majority, can lose. Industries with faster productivity growth have lost employment, but the losses have been offset by gains in customer industries and through higher final demand (Autor and Salomons 2018), and product-demand spillovers turned routine-replacing technical change in Europe into net job growth (Gregory, Salomons, and Zierahn 2022). Aggregate employment is therefore a weak test of either view; occupational employment, hiring, relative wages, and labor's share are more informative.

Finally, labor here is a single input. When automation removes an occupation's less expert tasks, wages in it rise and employment falls, whereas automation of its expert tasks lowers wages and raises employment (Autor and Thompson 2025), so employment counts alone can misstate the effect on workers. Equipment has been more substitutable for less skilled than for more skilled labor

(Krusell, Ohanian, Ríos-Rull, and Violante 2000), and σ for AI may likewise differ by experience—a possibility to which the evidence below returns.

Lessons from Past Automation

Before turning to AI, it is worth asking what the condition explains about earlier automation. Table 1 classifies twelve technology–occupation episodes on two observable dimensions. The first is whether the technology performed a subset of the incumbent occupation's tasks or its entire function, an indicator of whether σ was low or high. The second is whether output of the affected good or service grew faster than the productivity gain the technology delivered, an ex post indicator of elastic demand. Because this indicator is defined by total input, it is not independent of the employment outcome under partial automation, so the first regularity below is better read as a description than as a test. Elasticities were not estimated for most episodes, so the classification is qualitative; online Appendix B sets out the coding rules. The episodes were chosen for their prominence in debates about automation rather than sampled, so they illustrate mechanisms and do not measure how often each outcome occurs. The approach follows Atack, Margo, and Rhode (2019) in examining automation task by task. Figure 1 places ten of the episodes on the two dimensions.

Table 1

Twelve Episodes of Automation

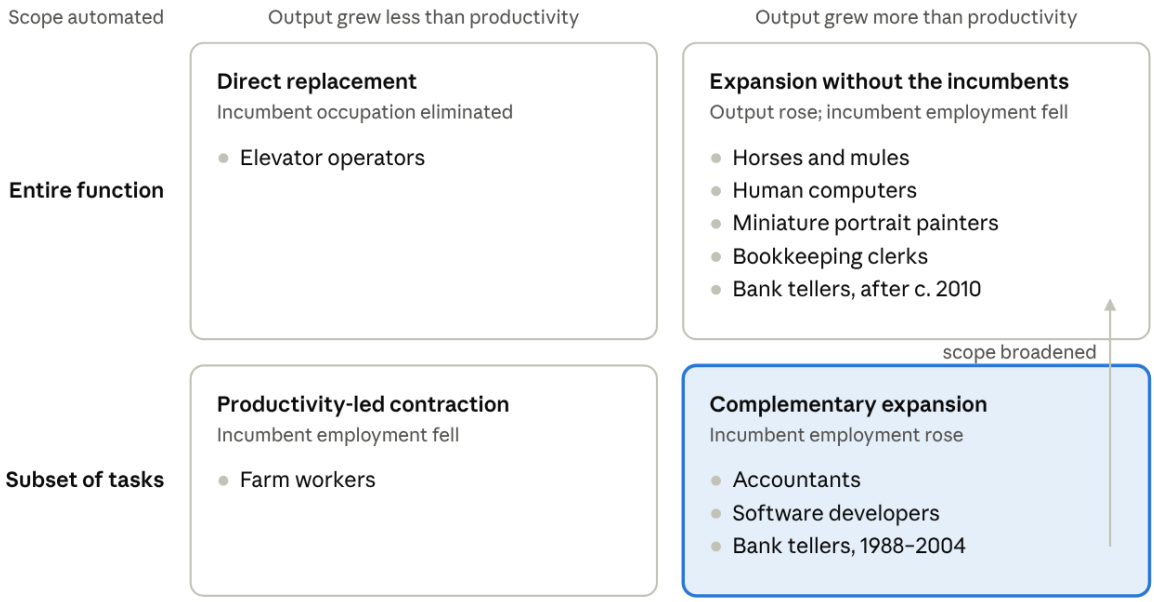
Technology (period)	Incumbent occupation	Scope automated	Demand for output	Incumbent employment	Source
Electronic spreadsheet (1980s–)	Accountants	Calculation, a subset of tasks	Expanded	Rose	Goldstein (2015)
Electronic spreadsheet (1980s–)	Bookkeeping and accounting clerks	Core function	Expanded	Fell	Goldstein (2015)
Automated teller machine (1988–2004)	Bank tellers	Cash handling, a subset of tasks	Expanded: urban branches rose 43 percent	Rose modestly nationally; tellers per urban branch fell from 20 to 13	Bessen (2015b); Autor (2015)
Online and mobile banking (c. 2010–)	Bank tellers	Most routine transactions	Transactions moved to digital channels	Falling; projected −13 percent over 2025–2035	BLS (2026d)
High-level languages, compilers, libraries (1957–)	Programmers and software developers	Writing code, a subset of tasks	Expanded	IT occupations rose tenfold, 1970–2014	Beckhusen (2016); online Appendix B
Deep-learning image analysis (2016–)	Radiologists	Image interpretation, a subset of tasks	Imaging volumes rising	Mayo Clinic radiologists up 55 percent since 2016, to more than 400	Lohr (2025)

Technology (period)	Incumbent occupation	Scope automated	Demand for output	Incumbent employment	Source
Electronic computers (1940s–1960s)	Human computers	Entire function (calculation)	Expanded	Occupation disappeared	Grier (2005)
Internal combustion engine (1900–1960)	Horses and mules	Entire function (traction)	Expanded	More than 21 million (1900) to about 3 million (1960)	Brynjolfsson and McAfee (2015)
Photography (1840s–1860s)	Miniature portrait painters	Entire function (recording a likeness)	Expanded	Many miniaturists abandoned the art after 1839	Barratt (2004); online Appendix B
Tractors and mechanization (1900–2000)	Farm workers	Power and harvesting, a subset of tasks	Grew less than productivity	From 41 to 2 percent of US employment	Autor (2015)
Automatic elevators (mid-twentieth century)	Elevator operators	Entire function	No induced demand	Occupation eliminated	Bessen (2016)
Automatic telephone switching (1920–1940)	Telephone operators	Entire function (connecting calls)	Not assessed	Incumbents displaced; later cohorts employed elsewhere	Feigenbaum and Gross (2024)

Source: Author's classification, based on the sources listed. *Note:* "Expanded" means that output of the good or service grew faster than the productivity gain the technology delivered.

Figure 1: Incumbents Gained Only When Automation Was Partial and Demand Grew

Ten episodes from Table 1, classified by scope of automation and demand response



Source: Author's classification of the episodes in Table 1. *Note:* Radiology is not placed because its employment trend cannot yet be attributed to AI; telephone switching is not placed because the demand response was not assessed.

Four Regularities

Subject to these caveats, four regularities emerge. The first is that partial automation in a growing market raised incumbent employment. This holds for accountants, for bank tellers during the spread of automated teller machines (ATMs), and for software developers. Bessen (2015a, 2015b) documents the mechanism for tellers: between 1988 and 2004, ATMs cut the number of tellers per urban bank branch from about 20 to 13, lowering the cost of operating a branch, and banks increased the number of urban branches by 43 percent, offsetting most but not all of the decline in tellers per branch. Nationally, teller employment rose modestly, from about 500,000 in 1980 to about 550,000 in 2010 (Autor 2015).

The second is that whole-function substitution reduced incumbent employment regardless of demand. This holds for horses, human computers, miniature painters, bookkeeping clerks, and elevator operators. The US horse and mule population grew sixfold between 1840 and 1900, to more than 21 million, and fell to about 3 million by 1960, although demand for transport rose throughout (Brynjolfsson and McAfee 2015); Leontief (1983) drew the analogy to human labor explicitly. In the language of the framework, these are cases in which σ was effectively infinite: demand for the output grew, and the growth was met entirely by the new input.

The third is that the elimination of whole occupations by automation has been rare. Of about 270

detailed occupations in the 1950 Census, Bessen (2016) finds only one, elevator operator, that was eliminated primarily by automation by 2010. Narrower job titles, such as the human computers in Table 1, did disappear, but census occupations bundle tasks that were automated at different times, and the bundles have usually survived.

The fourth, and the most relevant to AI, is that occupations migrate between cells as a technology's scope broadens. Teller employment held up during the spread of ATMs but has fallen since about 2010, as online and mobile banking took over routine transactions, and the Bureau of Labor Statistics projects a further 13 percent decline between 2025 and 2035 (BLS 2026d). A general-purpose technology that improves every year can move an occupation from partial to whole-function substitution after a period of complementarity. At the level of the whole economy, the creation of new work has repeatedly offset displacement, but Acemoglu and Restrepo (2019) find that the balance has shifted: in their decomposition, the displacement effect reduced the task content of production for labor by about 0.48 percent a year in 1947–1987 and 0.70 percent a year in 1987–2017, while the reinstatement effect raised it by about 0.47 and 0.35 percent a year.

Barely Cheaper Machines

The framework also explains why a technology's early years can be its worst for workers. Acemoglu and Restrepo (2019) call technologies that displace workers while raising productivity only a little "so-so technologies," with automated customer service among their examples; such technologies reduce labor demand because the productivity effect is too small to offset displacement. So-so automation is concentrated where machines first become competitive. When AI and labor are close substitutes in a task, the task moves to AI once AI's cost of performing it falls below labor's, that is, once $p_M/a_M < \mathrm{w}/a_L$. At that crossing point, displacement in the task is complete, but the cost saving—and with it the price reduction that drives demand expansion through ε—is close to zero. Each task that AI takes over at the margin of profitability therefore displaces labor with little offsetting expansion of demand; the offset grows only as AI's cost falls well below labor's in the tasks it has taken over. A technology that performs tasks at roughly the cost of labor is thus likely to look like so-so automation at first, even if its long-run productivity effects are large.

Radiology

Radiology is a test case chosen by a proponent of the substitution view. In October 2016, Hinton said that "people should stop training radiologists now," because within about five years deep learning would outperform them (Hinton 2016). Nine years later, the radiology staff of the Mayo Clinic had grown 55 percent, to more than 400 radiologists (Lohr 2025), and average US radiologist compensation reached $571,000 in 2025 (Medscape 2026). Hinton has since said that he spoke too broadly, that he meant image analysis specifically, and that he was wrong about timing but not about direction (Lohr 2025). Radiologists also consult, monitor patients, and perform procedures, so image interpretation is only part of the job—partial automation in the sense of Table 1. The case shows that the 2016 prediction was wrong about timing and about the scope of the job, but it does not identify a positive effect of AI on radiologists' employment. Imaging volumes were rising

before deep learning was deployed, the supply of radiologists is constrained by residency positions and reimbursement rules, and radiologists given AI predictions in an experiment did not use them well (Agarwal, Moehring, Rajpurkar, and Salz 2023). For these reasons radiology is not placed in Figure 1.

What Is Known about the Two Elasticities

No study has estimated either elasticity for generative AI. The available evidence is indirect, but it says a good deal about where each is likely to be high.

Demand for Cognitive Services

Demand that current prices exclude is the strongest case for a high ε. Low-income Americans received no legal help, or not enough help, for 92 percent of the civil legal problems that substantially affected their lives in the previous year (Legal Services Corporation 2022). Gaps of this kind indicate a large mass of potential demand at prices below current ones. They are consistent with a high ε over the relevant range of prices, but they are not elasticity estimates, because they do not measure willingness to pay at the prices AI would make feasible.

Satiation is the main argument on the other side. Bessen (2019) shows that in US textiles, steel, and motor vehicles, employment grew for decades alongside rapid productivity growth and then declined: demand was highly elastic when consumption was low and became inelastic as consumption approached satiation. Models of structural change formalize the same mechanism across sectors, through relative prices and through income effects on demand (Ngai and Pissarides 2007; Comin, Lashkari, and Mestieri 2021), and Autor (2015) emphasizes that the income elasticity of demand matters alongside the price elasticity. For AI, these results imply that ε may be high for cognitive services precisely because consumption of many of them is far below satiation, and that ε should be expected to fall as consumption rises.

Health care shows how far apart the two kinds of evidence can be. Hinton has argued that demand for health care is close to unlimited: "if you could make doctors five times as efficient, we could all have five times as much healthcare" (Hinton 2025b). Measured price elasticities point the other way. In the RAND Health Insurance Experiment, the elasticity of medical spending with respect to patients' out-of-pocket price was approximately -0.2 (Manning et al. 1987). That estimate is not the market elasticity relevant here, because insurance mutes the pass-through of costs to patients, and it does not capture rationing through waiting times and scarce providers, which cheaper provision could relieve. The case for elastic demand in health care rests on the removal of non-price rationing, not on measured price elasticities. Hinton himself argues that most occupations lack this property—"most jobs, I think, are not like that"—and expects a person working with an AI assistant to do work that once took ten people, with paralegals and call-center workers displaced early (Hinton 2025b).

The evidence is thus consistent with $\varepsilon > 1$ in markets where current prices or non-price rationing exclude much of the potential demand, although no study measures it there. It does not support

a general presumption of elastic demand for cognitive work, and the historical record implies that elasticities fall as markets mature. Even where $\varepsilon > 1$, employment rises only if ε also exceeds σ.

Substitutability of AI for Labor

The literature on the elasticity of substitution between capital and labor provides a benchmark, and it is contested. Karabarbounis and Neiman (2014) estimate an elasticity of about 1.25 and attribute roughly half of the global decline in the labor share to the falling relative price of investment goods. Microeconomic estimates are lower: Oberfield and Raval (2021) estimate 0.5 to 0.7 for US manufacturing, and a meta-analysis of more than 3,000 estimates finds that correcting for publication bias and method choices lowers the mean from about 0.9 to about 0.3 (Gechert, Havranek, Irsova, and Kolcunova 2022).

These estimates concern equipment and structures, which perform few tasks from start to finish, and there is a specific reason to expect AI to be more substitutable. Earlier computerization was confined to tasks whose rules could be written down. It struggled with tasks that people perform without being able to say how—Polanyi's (1966) observation that "we can know more than we can tell." Autor (2015) discussed this constraint, which he called Polanyi's paradox, as a central limit on the substitution of computers for workers, and noted that machine learning, which infers rules from examples, potentially circumvents it, although he doubted that it would do so soon. Generative AI has done so for a wide range of language and image tasks, so the constraint that kept σ low for earlier information technology binds less for AI.

Task-level evidence suggests that σ varies widely across tasks and depends on complementary investment. Eloundou, Manning, Mishkin, and Rock (2024) estimate that roughly 1.8 percent of US jobs could have over half their tasks affected by large language models with simple interfaces, and just over 46 percent once software built to complement the models is taken into account; the gap between the two figures indicates how much of AI's potential reach depends on complementary investment. In a field experiment with 758 consultants, access to GPT-4 raised performance on tasks within the model's capabilities—by 43 percent for consultants below the average on a baseline assessment and by 17 percent for those above it, in the working-paper version (Dell'Acqua et al. 2023)—but made consultants 19 percentage points less likely to reach a correct answer on a task just outside those capabilities that seemed similar in difficulty (Dell'Acqua et al. 2026). In an experiment with radiologists, AI predictions did not improve average diagnostic performance, because radiologists did not weight them appropriately; the best-performing design assigned each case either to the AI or to the radiologist alone (Agarwal, Moehring, Rajpurkar, and Salz 2023). That is substitution case by case, not collaboration. Agrawal, Gans, and Goldfarb (2019) offer a framework for the pattern: cheaper prediction complements the human judgment applied to a decision, but it can also make it profitable to automate the decision altogether.

In markets with few frictions, the net effect on human work has been negative and fast, which suggests that substitution dominates there. After ChatGPT's release, freelancers on a large online platform in exposed occupations saw monthly jobs fall by about 2 percent and monthly earnings by about 5 percent, and higher-quality freelancers were not protected (Hui, Reshef, and Zhou

2024). Posts for writing and coding jobs on a freelancing platform fell 21 percent relative to manual-intensive jobs within eight months (Demirci, Hannane, and Zhu 2025). Platform markets, with discrete tasks and short contracts, are likely to show the highest substitution, so these estimates are better read as upper bounds on displacement than as typical values.

Technical feasibility also overstates economic substitution. For computer vision, Svanberg et al. (2024) estimate that only 23 percent of the worker compensation in tasks exposed to the technology would be cost-effective for firms to automate at current costs, because of the large up-front costs of AI systems. Exposure measures identify where σ could be high; deployment costs determine where it is high at today's prices. The evidence as a whole points to an elasticity that varies widely across tasks and workers: high for discrete tasks within the frontier of AI capabilities, low outside it, and limited everywhere by the cost of putting AI into production.

Evidence since ChatGPT

This section reviews evidence on AI and employment from the release of ChatGPT in November 2022 through mid-2026; online Appendix C summarizes the main studies in a common format, ordered by research design. Most are working papers or policy reports and remain subject to revision. They also measure exposure to AI differently. Some use assessments of what AI could technically do in each occupation (Felten, Raj, and Seamans 2021; Eloundou, Manning, Mishkin, and Rock 2024), building on earlier measures that emphasized AI's reach into high-skill tasks (Webb 2020); others use observed AI use. Exposure measures describe what AI could do, not the elasticities that govern employment, so a comparison of exposed with unexposed occupations estimates the net of substitution and demand expansion, not either one alone.

Aggregate Outcomes

Studies of aggregate outcomes find no detectable effect. A synthetic difference-in-differences analysis of US monthly microdata finds effects on the employment shares and real wages of exposed occupations that are not statistically distinguishable from zero (Nunn 2026); earlier tracking found no discernible disruption in the 33 months after ChatGPT's release (Gimbel, Kinder, Kendall, and Lee 2025). Since 2022, unemployment has risen 0.77 percentage points for the most exposed quintile of US workers and 0.85 points for the least exposed (Mahoney, McEntarfer, and Wahal 2026), and Massenkoff and McCrory (2026) find no systematic increase in unemployment among highly exposed workers. In Danish administrative data linked to a survey of about 25,000 workers in exposed occupations, Humlum and Vestergaard (2026) rule out effects on earnings or hours larger than 2 percent two years after ChatGPT's release. Across nearly 6,000 executives in the United States, the United Kingdom, Germany, and Australia, more than 90 percent on average report that AI has had no impact on their firm's employment over the past three years (Yotzov et al. 2026).

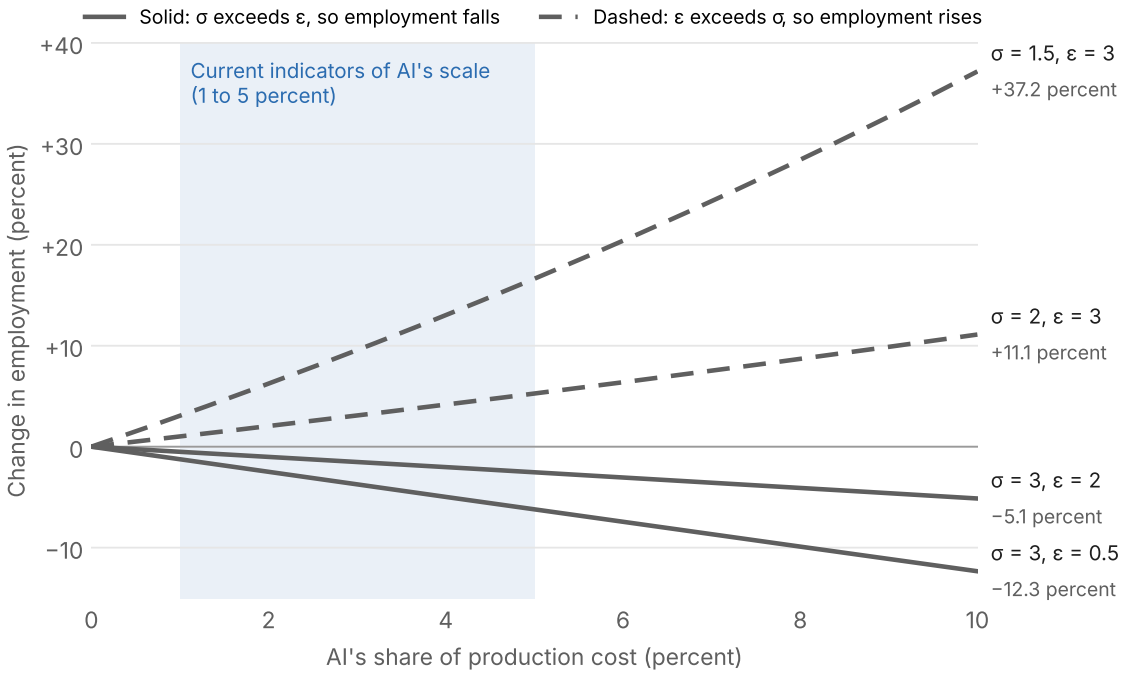
Why the Null Results Were to Be Expected

These null results are weak evidence on the question the two views dispute. The derived-demand condition scales the employment effect of cheaper AI services by AI's share of costs, and the effect of copilots by the productivity gains they deliver, and both remain small. No statistic measures AI's cost share directly, but the indicators point the same way. Between December 2025 and May 2026, 17 to 20 percent of US businesses reported using AI in some business function in the previous two weeks (Grundy, Breaux, and Khatiwoda 2026). Generative AI assisted between 1 and 5 percent of all US work hours in late 2024 (Bick, Blandin, and Deming 2026), and the hours it saved rose from 1.6 percent of all work hours in the third quarter of 2024 to 2.2 percent in the second quarter of 2026 (Federal Reserve Bank of St. Louis 2026). In Danish data, workers who used chatbots reported saving about 2.8 percent of their work hours, and only 3 to 7 percent of those productivity gains passed through to their earnings (Humlum and Vestergaard 2025). Acemoglu (2025) estimates that AI will raise total factor productivity by no more than 0.66 percent over ten years.

Figure 2 shows what magnitudes like these imply. Because the effect of a large price decline accumulates as AI's cost share rises, the cumulative change in industry employment equals $(\sigma - \varepsilon)/(\sigma - 1)$ times the change in the log of labor's cost share (online Appendix A). At a cost share of 5 percent—the top of the 1 to 5 percent range of work-hour shares above, and above the 2.8 percent time saving reported by Danish users— three of the four combinations of elasticities in the figure imply employment changes of about 6 percent or less in either direction, accumulated over the several years it takes the share to rise. AI's share is presumably higher in the most exposed occupations and industries, which is where effects should appear first. Larger effects require a cost share well above current levels, an elasticity of substitution close to one, or a demand elasticity far above σ; with σ close to one, output can expand a great deal before AI's share rises much. The aggregate null results rule out large short-run disruption; they say little about the sign of $\varepsilon - \sigma$.

Figure 2: Employment Effects Are Bounded by AI's Cost Share

Change in industry employment implied by AI's share of production cost, for four combinations of σ and ε



Source: Author's calculations. *Note:* Each line plots the cumulative change in industry employment, $\exp\{[(\sigma-\varepsilon)/(\sigma-1)]\cdot\ln(1-s_M)\}-1$, as AI's cost share rises from zero to s_M, holding the wage, labor productivity, and the frontier of AI capabilities fixed (online Appendix A). All four cases have $\sigma > 1$, because only then does a fall in AI's effective price raise its cost share; with $\sigma = 1$ the share is constant, and the effect equals $s_M(1-\varepsilon)$ times the log change in AI's effective price. The shaded band spans 1 to 5 percent, the range of indicators of AI's current scale in production (shares of work hours assisted by or saved with generative AI), which are not direct measures of AI's cost share.

That AI's share has stayed small after a thousandfold fall in the list price of inference is itself informative. The ratio of AI's cost share to labor's moves with their relative effective prices raised to the power $1-\sigma$ (online Appendix A), so if σ were well above one and the effective price of AI services had fallen as far as list prices, AI's share would have risen steeply: with $\sigma = 2$, for example, a thousandfold fall in AI's effective price would raise a 0.1 percent cost share to about 50 percent, and with $\sigma = 1.5$, to about 3 percent. Three explanations, not mutually exclusive, reconcile the facts. The effective price of AI services—per task completed, including the costs of integration, verification, and errors—may have fallen far less than the price per token; the gap between exposure estimates with and without complementary software, and task-level estimates of deployment costs, both point that way. Adoption may lag behind prices. Or σ for current capabilities may be close to one, in which case AI's share would rise little however far its price fell. The first two explanations imply that larger effects lie ahead as complementary investment accumulates. The third does not rule them out: with σ near one, employment effects are not bounded by AI's share, so the

null results would then also imply either that ε is close to one or that the effective price has fallen far less than list prices.

The pattern has a precedent. Establishments exposed to an earlier generation of AI reduced their non-AI hiring, but effects at the level of occupations and industries were too small to detect (Acemoglu, Autor, Hazell, and Restrepo 2022). Aggregate stability can also conceal offsetting changes, as the housing boom masked the decline in US manufacturing employment in the 2000s (Charles, Hurst, and Notowidigdo 2016).

The Hiring Margin

Labor-saving technology often reduces employment through fewer hires rather than through separations. In Germany, industrial robots did not raise incumbent workers' risk of job loss; manufacturing employment fell because fewer young workers entered it, and service employment fully offset the decline (Dauth, Findeisen, Suedekum, and Woessner 2021). When automatic switching replaced US telephone operators, incumbents were more likely to be in lower-paying jobs or out of work a decade later, but later cohorts of young women were absorbed by clerical and service jobs (Feigenbaum and Gross 2024). Automation does harm some incumbents (Bessen, Goos, Salomons, and van den Berge 2025), but an effect concentrated on hiring is invisible in occupation-specific unemployment rates, because a worker never hired into an exposed occupation is not counted as unemployed from it. The stakes are not small: workers who enter the labor market in poor conditions suffer persistent losses (von Wachter 2020).

Several US data sources show a decline on this margin. In ADP payroll data, as of June 2026 employment of 22- to 25-year-olds in the two most AI-exposed quintiles of occupations was 19 percent below the level it would have reached had it grown in step with that of the same age group in the three least exposed quintiles, up from 15 percent a year earlier; the decline operates through hiring rather than separations, although the regression estimate roughly halves when the college share of each occupation is controlled for (Brynjolfsson, Chandar, and Chen 2025). Census data covering more than 95 percent of private-sector jobs show early-career employment in the most exposed industries falling about 12 percent relative to other industries over ten quarters after ChatGPT's release, with monetary policy explaining at most a quarter of the gap, although hiring had largely recovered by early 2025 (Tucker 2026). Graduates of the most exposed decile of college majors became 5 percentage points less likely to find employment soon after graduation and earned about 13 percent less initially (Orr, Tucker, and Warren 2026), and junior employment at firms that adopted generative AI fell by roughly 8 to 10 percent relative to other firms, mainly through slower hiring, while senior employment kept rising (Hosseini Maasoum and Lichtinger 2025). The aggregate magnitude is modest: in the Current Population Survey, the decline is confined to workers aged 20 to 24 and amounts to about 0.1 percentage point of aggregate unemployment (Atkinson and Yamco 2026).

Other data show no comparable effect. Unemployment among US college graduates aged 22 to 25 averaged 7.3 percent in summer 2026, within its 2022–2025 range, and did not rise relative to that of older graduates (Fairlie and Wu 2026). Administrative data from Finland show no displacement

of young workers linked to AI exposure (Kauhanen and Rouvinen 2026). Norwegian register data show none through March 2025 (Facius and Iacono 2026) and, through February 2026, only an emerging gap for 22- to 25-year-olds that the authors describe as a pattern rather than an identified effect (Hernæs and Kostøl 2026). Danish workplaces that adopted chatbots show no differential change in employment or in the composition of hires (Humlum and Vestergaard 2026).

Attribution

The US decline in entry-level employment in exposed occupations thus appears in several independent data sources, but its attribution to AI is not established. Unemployment risk in AI-exposed occupations began rising in early 2022, months before ChatGPT's release, and entry into these occupations fell from the 2021 graduating cohort onward (Frank et al. 2026). Junior hiring weakened as interest rates rose from 2022. With firm-by-time controls, the relative decline becomes significant only in 2024 and is not explained by differences in interest-rate sensitivity (Brynjolfsson, Chandar, and Chen 2026), but the firm-level estimates proved sensitive to changes in data processing in the August 2026 revision of Brynjolfsson, Chandar, and Chen (2025). In résumé and job-posting data for the United States, the United Kingdom, Canada, and Australia, controlling for exposure to remote work sharply attenuates the estimated effect of AI exposure on junior hiring, often to statistical insignificance (Lambert and Schindler 2026), although in US payroll data the same control changes the estimate little (Brynjolfsson, Chandar, and Chen 2025). Until these issues are resolved, the entry-level evidence is consistent with the substitution view but does not establish it.

Adopting Firms and Software Development

Firm-level evidence points toward complementarity, subject to two caveats. Firms with the most intensive AI spending increased total headcount by about 10 percent and entry-level headcount by about 12 percent in the two years after adoption, relative to later adopters with similar spending intensity (Kharazian, Simon, and Stevens 2026). Headcount is measured from online professional profiles, the authors present the result as an association, and the high-intensity group shows some pre-trends. More fundamentally, a firm's demand elasticity includes business taken from rivals, so expansion at adopters is consistent with $\varepsilon < \sigma$ for the industry as a whole. Hampole, Papanikolaou, Schmidt, and Seegmiller (2025) find that AI exposure reduces demand for exposed tasks within firms but that productivity-driven expansion offsets much of the reduction—the two terms of the derived-demand condition, measured within firms.

Software development, where AI coding agents are most widely deployed, shows a shift in the composition of demand. US software-development postings on Indeed rose almost 15 percent between February 2025 and mid-2026 while overall postings fell about 7 percent, although software postings remained about 27.5 percent below their February 2020 level; senior roles accounted for 71 percent of the increase (Gallacher 2026). Entry-level roles made up 4.5 percent of software-development postings in the first quarter of 2026, the lowest share of any sector, and across all occupations, senior postings rose 14.7 percent in the year to May 2026 while entry-level postings fell 7.5 percent (Aidala and Puri 2026). This is the pattern the framework predicts if σ differs

by experience, with AI substituting for the tasks assigned to entrants and complementing those performed by experienced workers. It is also consistent with a cyclical tilt toward experienced hires under uncertainty, which the data cannot yet distinguish.

In sum, the evidence through mid-2026 shows no aggregate employment effect, a decline in US entry-level employment in exposed occupations that appears in several data sources but is not yet attributed to AI, and a shift in labor demand toward experienced workers. Because AI's cost share is still small, none of these findings discriminates sharply between the two views.

Why Novices Gain Most and Are Hired Least

Field experiments with generative AI have consistently found the largest productivity gains for less experienced or less skilled workers. Customer-support agents given an AI assistant became about 15 percent more productive on average, with gains of about 30 percent for less experienced and less skilled agents and minimal gains for the most skilled (Brynjolfsson, Li, and Raymond 2025). In a writing experiment, ChatGPT reduced time per task by 40 percent, raised quality by 18 percent, and narrowed the gap between stronger and weaker writers (Noy and Zhang 2023). Among consultants, the gain in performance was more than twice as large for those below the average on a baseline assessment as for those above it (Dell'Acqua et al. 2023). In field experiments with nearly 5,000 software developers at three companies, access to a coding assistant raised the number of completed tasks by about 26 percent, with larger point estimates for junior developers, although the difference by experience is not statistically significant (Cui et al. 2026).

Read through the copilot channel, these results do not necessarily favor junior hiring. Applying the labor-augmenting formula to junior workers, with cost share s_J and elasticity of substitution σ_J between junior labor and the other inputs, junior employment rises with juniors' productivity if $s_J(\varepsilon - 1) + (1 - s_J)(\sigma_J - 1) > 0$ (online Appendix A). Because juniors' cost share is small, this requires roughly $\sigma_J > 1$: juniors must substitute readily for other inputs, including the time of experienced workers. If instead juniors complement experienced workers ($\sigma_J < 1$), a productivity gain concentrated among juniors reduces the number a firm needs. That is one possible resolution of the puzzle posed by the hiring data, in which entry-level employment in exposed occupations shows the clearest declines.

The framework suggests a second resolution. The experiments measure the copilot channel: they hold employment fixed and estimate how much more a worker produces with AI. Hiring also responds to the agent channel, in which AI performs tasks itself at a falling price. Through that channel, junior employment responds to the effective price of AI services in proportion to $s_M(\sigma_{JM} - \varepsilon)$, where σ_{JM} is the elasticity of substitution between junior labor and AI services (online Appendix A). The two channels are governed by different parameters, and both can operate at once: AI can raise the productivity of the juniors a firm employs while reducing the number it needs to hire.

There is a reason to expect the two to go together. Novices gain most from AI because it supplies

what they lack: codified knowledge and the accumulated practices of experienced workers, and Brynjolfsson, Li, and Raymond (2025) present suggestive evidence that their assistant disseminated such practices from the most productive agents to less experienced ones. A system that can supply an expert's practices to a novice can often apply them without the novice, so the capability that raises juniors' productivity through the copilot channel also raises σ_{JM} through the agent channel. Ide and Talamàs (2025) formalize a related asymmetry in a model of the knowledge economy: nonautonomous AI, which acts solely as a copilot, benefits the least knowledgeable individuals, while autonomous AI primarily benefits the most knowledgeable. And when automation removes an occupation's less expert tasks, wages in it rise while employment falls (Autor and Thompson 2025); the less expert tasks are disproportionately those assigned to entrants.

This resolution has implications that distinguish it from two alternatives: that the decline in junior hiring has causes unrelated to AI, and that copilots themselves reduce the number of juniors firms need because juniors complement other inputs ($\sigma_J < 1$). If the agent channel drives the decline, junior hiring should fall most where firms deploy AI to perform tasks on its own, through programming interfaces and agents, rather than where workers use it as an assistant. Within firms, it should fall in the teams and tasks where agents are deployed, and the gap between junior and senior employment should widen with spending on autonomous deployment, holding total AI spending fixed. Junior hiring that falls instead with spending on per-seat assistants would favor the copilot explanation. Occupation-level evidence already points this way. Classifying occupations by whether AI use in their tasks, as recorded in the Anthropic Economic Index, mostly automates or mostly augments them, Brynjolfsson, Chandar, and Chen (2025) find that employment of 22- to 25-year-olds fell most in the occupations with the highest automation shares, holding augmentation and overall usage fixed, while employment was flat or rising in the occupations with the highest augmentation shares. The classification describes how one provider's users engage with AI rather than how firms deploy it, and the authors present the pattern as descriptive, so the within-firm tests remain the sharper ones. Usage data from one AI provider indicate that about three-quarters of firms' use through programming interfaces involves delegating tasks rather than seeking assistance (Appel et al. 2025). Experiments that randomize access to agents across teams, and measure staffing rather than output per worker, would identify σ_{JM} directly.

The answer matters beyond the static framework. If entry-level hiring stays depressed, the supply of experienced workers will fall with a lag, because experienced workers are produced from entrants. The demand-expansion view relies on experienced workers who complement AI; fewer entrants today would mean fewer such workers a decade from now.

Measuring the Two Elasticities

Both elasticities are theoretical objects, and a common objection is that they cannot be measured for a technology this new. They can, because each corresponds to an observable response to the same shock. When the effective price of AI services changes, unit cost changes by, to a first approximation, $s_M \cdot \Delta \ln(p_M/a_M)$; call the actual change in log unit cost Δln p, which with full pass-through is

also the change in the output price. Employment then changes by

$$\Delta \ln L \;=\; \underbrace{-\,\varepsilon\,\Delta \ln p}_{\text{scale effect}} \;+\; \underbrace{\sigma\,\Delta \ln p}_{\text{substitution effect}}\;, \qquad \Delta \ln p \approx s_M\,\Delta \ln(p_M/a_M).$$

The first term is the change in output, Δln Q; the second is the change in labor per unit of output, Δln(L/Q). Each elasticity is therefore a ratio of observable changes to the same cost reduction: $\varepsilon = -\Delta$ln Q/Δln p and $\sigma = \Delta$ln(L/Q)/Δln p (online Appendix A). These ratios hold with wages and labor productivity fixed; otherwise σ should be computed against the change in unit cost relative to the effective wage. A worked example shows the logic. Suppose a fall in the effective price of AI services lowers an industry's unit cost, and with full pass-through its output price, by 3.5 log points—to a first approximation, what halving the price of an input with a 5 percent cost share would do. If output rises by 10 log points and labor per unit of output falls by 15 log points, then $\varepsilon \approx 10/3.5 \approx 2.9$, $\sigma \approx 15/3.5 \approx 4.3$, and employment falls by 5 log points. Demand in this industry is highly elastic, yet employment falls.

Three measurement choices determine whether such estimates are informative. The price must be the effective price of AI services—the cost per task completed at a given quality, including integration, verification, and the cost of errors—rather than the list price per token, and the earlier discussion suggests that the two have diverged widely. Output and prices must be measured for the market, not the firm, because a firm's output responds to business taken from rivals. And the cost reduction must be unrelated to other shocks to the industry. Shift-share designs that combine an industry's or occupation's predetermined exposure to particular AI capabilities with the timing of model releases and price cuts are a natural approach (Borusyak, Hull, and Jaravel 2025), but the attribution debate reviewed above shows that exposure to AI is correlated with sensitivity to interest rates and to remote work, so such designs need to control for those shocks.

A second route uses data on spending rather than output. With a constant elasticity of substitution, the ratio of AI's cost share to labor's satisfies $\Delta\ln(s_M/s_L) = (1-\sigma)\cdot\Delta\ln[(p_M/a_M)/(\mathrm{w}/a_L)]$ (online Appendix A), so σ can be estimated from how AI's share of costs responds to its relative effective price across firms or industries. Much of the capital–labor literature proceeds this way; Oberfield and Raval (2021), for example, use local variation in wages to estimate plant-level elasticities. Firm expense data of the kind used by Kharazian, Simon, and Stevens (2026), which record spending on AI services alongside headcount, make the approach feasible for AI. Its weaknesses are the mismeasurement of effective prices and the lags with which firms adjust.

A third route works task by task. Where AI and labor are close substitutes in a task, the relevant object is not a finite σ but the cost at which the task moves from labor to AI: the task is automated when the effective cost of AI falls below that of labor. Svanberg et al. (2024) apply this logic to computer vision, combining task inventories, wages, and engineering estimates of system costs. Applied to generative AI with costs measured per completed task, the approach traces out how much of an occupation's wage bill crosses the threshold as AI's effective price falls. The slope of that schedule is the counterpart of σ in task-based models, where displacement operates at the

extensive margin (Acemoglu and Restrepo 2018), and the approach can be implemented before the economic data arrive.

A fourth route, for ε alone, is in one respect the easiest. The demand elasticity is a property of product demand, not of AI, so it can be estimated from any cost shock to the industry—changes in input prices, regulation, or taxes—using the methods of the pass-through literature, without waiting for AI's effects to appear. At the level of the firm, markups provide a bound: under monopolistic competition, a firm's demand elasticity equals $\mu/(\mu - 1)$, where μ is its markup, which can be estimated from production data (De Loecker and Warzynski 2012). A markup of 1.25 implies a firm-level elasticity of 5. Because firm-level elasticities include business stealing, they bound the market elasticity from above, and where an industry's firm-level elasticity is below plausible values of σ, the demand-expansion view cannot hold. Where demand is rationed by waiting times or scarce providers, as in health care, price elasticities understate the demand that cheaper provision could release, and the quantities served at given prices are the better guide.

Table 2 summarizes the four routes. None requires new theory. They require effective prices of AI services, which no statistical agency yet collects, and AI's share of costs by industry, which business surveys could measure directly by asking firms what they spend on AI services.

Table 2

Four Routes to Measuring ε and σ for AI

Route	What it identifies	Data required	Main threat
Responses of output and labor input to AI cost shocks	$\varepsilon = -\Delta\ln \mathrm{Q}/\Delta\ln \mathrm{p}$ and $\sigma = \Delta\ln(\mathrm{L}/\mathrm{Q})/\Delta\ln \mathrm{p}$, where $\Delta\ln$ p is the change in unit cost, approximately $s_M \cdot \Delta\ln(p_M/a_M)$	Industry output, prices, and employment; AI's cost share; effective prices of AI services; exposure interacted with model releases or price cuts	Exposure correlated with other shocks, such as interest rates or remote work; measurement at the firm rather than the market
Cost-share equation	σ, from $\Delta\ln(s_M/s_L) = (1 - \sigma)\cdot\Delta\ln$ of relative effective prices	Spending on AI services and payroll by firm or industry; quality-adjusted prices of AI services; wages	Mismeasured effective prices; adjustment lags
Task-level cost comparisons	Share of an occupation's wage bill for which AI is cheaper than labor, and how it moves with AI's price	Task inventories; wages; cost of AI per completed task, including integration and error correction	Engineering estimates may omit organizational costs; the frontier of capabilities moves
Pass-through and markups	ε from any cost shock to the industry; firm elasticity $\mu/(\mu - 1)$ as an upper bound	Industry prices and quantities; firm markups estimated from production data	Business stealing in firm-level elasticities; non-price rationing

Source: Author's summary. *Note:* p is the output price, Q output, L employment, s_M and s_L the cost shares of AI services and labor, p_M/a_M the effective price of AI services, w/a_L the effective wage, and μ a firm's markup. Derivations are in online Appendix A.

Will Human Tasks Persist, and Will New Ones Appear?

The two views differ chiefly in their forecasts of σ—or, in task-based terms, in whether the set of tasks in which labor holds a comparative advantage stays large as AI improves and is replenished by new tasks. Korinek and Suh (2024) formalize the alternatives. If the complexity of tasks is unbounded and automation proceeds gradually, wages can rise indefinitely; if the tasks humans can perform are bounded and automation reaches the boundary, wages collapse, and they can fall even earlier if automation outpaces the accumulation of capital. Trammell and Korinek (2026) survey the growth implications of such scenarios.

Four mechanisms support persistence. First, when tasks are complements, growth is constrained by the tasks that are essential and hard to improve, and value accrues to the inputs that perform them (Baumol 1967; Aghion, Jones, and Jones 2019; Jones 2026). With perfect complements, for example, automating tasks that took half of all labor time at most doubles output, however cheap the automated tasks become. When production requires the error-free completion of every step, raising reliability at some steps raises the value of reliability at the others (Kremer 1993): if AI drafts most of a legal brief, the value of the final human review can rise. Second, AI services require compute, energy, and chips in limited supply, and in Restrepo's (2025) model, scarce compute is directed to bottleneck work that is essential for growth, and some supplementary work may be left to people even though AI could perform it. Third, clients, regulators, and courts require a person to bear liability for audits, diagnoses, and engineering designs; much bottleneck work is physical—Hinton's advice to young people is to "train to be a plumber" (Hinton 2025b)—and consumers pay for human performance even where machines are superior, as in athletics. Fourth, substitutability is partly chosen. Research priorities and tax policies that favor human-like AI raise its substitutability for labor (Brynjolfsson 2022), so different priorities could lower it.

Several considerations work against persistence. Capability forecasts by some of the field's leaders imply that the set of human-advantaged tasks is small and shrinking: in 2023, Hinton estimated that AI surpassing human intelligence could arrive within 5 to 20 years (Hinton 2023), and in 2025 he said there was a good chance it would arrive within ten years (Hinton 2025a). Comparative advantage guarantees labor a use only at some wage, and in any task where AI is a perfect substitute, that wage cannot exceed the cost of the AI services that perform it; Restrepo (2025) shows that once all bottleneck work is automated, wages converge to the cost of the compute needed to reproduce human work, although in his model total wages exceed those paid before automation even as labor's share of income goes to zero. Compute scarcity is transitory: investment of hundreds of billions of dollars a year is directed at removing it, and the price of inference has fallen rapidly (Demirer, Fradkin, and Tadelis 2026). And accountability scales, because one accountable professional can sign off on the output of many AI systems. Preferences, likewise, protect activities rather than employment: horse riding survived the automobile as recreation, while horse-drawn transport did not. And the incentive to automate rises with wages (Hémous and Olsen 2022), so success in the tasks that remain human invites their automation.

The first set of mechanisms shows that persistence is possible; the second, that it is not guaranteed.

Which set dominates depends on how broadly and how quickly AI capabilities improve, and that depends on incentives and policy as well as on computer science. The derived-demand framework describes how capabilities translate into employment; it cannot forecast the capabilities. On this question, neither view can yet claim support from the data. Korinek and Juelfs (2022) discuss how policy might respond if wages were to fall.

Whether today's human tasks persist is only half of the question. The other half is the second term of the extended condition: whether new tasks for people keep appearing. Historically they have. About 60 percent of US employment in 2018 was in job titles that did not exist in 1940, and new work emerged where innovations complemented the outputs of occupations, while innovations that automated tasks slowed its emergence (Autor, Chin, Salomons, and Seegmiller 2024). Rapid change may itself create such work, since each new capability has to be built into products, connected to existing systems, verified, governed, and taught. But reinstatement has slowed relative to displacement since the late 1980s (Acemoglu and Restrepo 2019), and the contest that matters now is between the creation of new tasks for people and AI's absorption of them. New tasks need not remain human forever to sustain employment—only long enough, and in sufficient volume, to offset displacement elsewhere—but a technology that learns tasks from examples may absorb new ones faster than any before it. And when new work arrives, it tends to be taken up by new cohorts rather than by displaced workers, as when automatic switching replaced telephone operators (Feigenbaum and Gross 2024).

Who Gains, and When?

Over long horizons, productivity growth has raised incomes across the distribution. Nearly 80 percent of the world's population lived in extreme poverty in 1820; by 2019 the share had fallen below 10 percent (Roser 2016). In the United States between 1979 and 2021, average household income grew in every quintile, most in the highest, and transfers and taxes generated more income growth at the bottom than would otherwise have occurred (Congressional Budget Office 2024; Habib 2024). Lower prices for services such as legal assistance, tutoring, and medical advice would raise real incomes even with unchanged wages.

Those gains reached workers mainly through wages. The framework does not determine wages, but it implies that labor's share of the gains falls when $\sigma > 1$: with constant-elasticity technology, a fall in the effective price of AI services then raises AI's share of costs and lowers labor's. Karabarbounis and Neiman (2014) attribute part of the global decline in the labor share to the same mechanism operating through cheaper investment goods, and the labor share of output in the US nonfarm business sector fell to 52.8 percent in the second quarter of 2026, the lowest level in data going back to 1947, a decline that long predates generative AI (BLS 2026b; see Karabarbounis 2024 for a review of explanations). Automation that raises the return to wealth can widen inequality beyond what wages alone imply (Moll, Rachel, and Restrepo 2022). A falling labor share is nonetheless compatible with a rising average wage (Caselli and Manning 2019; Restrepo 2025). Hinton's forecast that AI "will make a few people much richer and most people poorer" (quoted

in Ma 2025) therefore implies more than $\sigma > 1$: it requires the wages of most workers to fall, which can happen in tasks where AI is a perfect substitute or when workers differ in the tasks they can perform.

Even when labor remained complementary to capital, wage gains lagged output gains for decades. In Britain between 1780 and 1840, output per worker rose 46 percent while real wages rose 12 percent; between 1840 and 1900, output per worker rose 90 percent and real wages 123 percent (Allen 2009). The movement of labor out of US agriculture, from 41 percent of employment in 1900 to 2 percent in 2000, spanned more than two generations (Autor 2015). A transition that is benign in the long run can be costly for the cohorts that live through it. Unlike horses, workers can own capital and vote (Brynjolfsson and McAfee 2015), so the distribution of the gains from AI will depend on ownership, taxation, and transfers as well as on technology. This article does not evaluate specific policies.

Pre-specified Predictions

The predictions and thresholds below were fixed in October 2026, before the outcome data exist. Pre-specification has costs as well as benefits (Olken 2015; Coffman and Niederle 2015); here its purpose is to make a forecast about a contested question checkable. Both predictions are my own forecasts, and the demand-expansion view implies both.

Prediction 1 (*subjective probability 0.7*). The 2031 annual average employment-to-population ratio of US residents aged 25 to 54 will be at or above its 2025 annual average (Bureau of Labor Statistics series LNS12300060, from the Current Population Survey). This prediction tests for large-scale disruption rather than for the sign of $\varepsilon - \sigma$.

Prediction 2 (*subjective probability 0.6*). Total employment in the occupations in the highest quintile of AI exposure will be higher in May 2031 than in May 2025. Exposure is the GPT-4-rated β measure of Eloundou, Manning, Mishkin, and Rock (2024), aggregated from O*NET-SOC codes to six-digit Standard Occupational Classification codes as an unweighted mean; quintiles contain equal shares of May 2025 employment; employment is measured with the Occupational Employment and Wage Statistics. If the Standard Occupational Classification is revised before 2031, occupations will be mapped with the official crosswalk.

Both predictions are joint tests of AI's effects and of macroeconomic conditions: a recession in 2031 could falsify them for reasons unrelated to AI. The first four indicators in Table 3 compare more with less exposed workers or firms and so difference out aggregate conditions; the last two are aggregate series that serve as corroborating evidence. They will be evaluated with data through the end of 2028.

Table 3

Indicators That Would Discriminate between the Two Views by the End of 2028

Indicator	Measure and source	Evidence for substitution if	Evidence for demand expansion if
Experienced workers	Employment of workers over age 30 in the two most AI-exposed quintiles of occupations relative to the three least exposed quintiles, ADP data (Brynjolfsson, Chandar, and Chen 2025)	Falls 5 percent or more behind peers, relative to late 2022	Employment and relative wages hold or rise
Entry-level workers	The same measure for ages 22–25 (19 percent as of June 2026), checked against estimates with firm-by-time and interest-rate controls (Brynjolfsson, Chandar, and Chen 2026)	Gap widens beyond 25 percent	Gap narrows below 10 percent
AI-adopting firms	Headcount and revenue of adopters relative to comparable nonadopters, in designs that address selection into adoption and business stealing	Revenue grows while headcount shrinks	Headcount grows with revenue
Agents versus copilots	Junior and senior headcount at adopting firms by type of AI spending—programmatic access and agents versus per-seat assistants—from firm expense data linked to headcount, as in Kharazian, Simon, and Stevens (2026)	Junior headcount falls with spending on agents, holding total AI spending fixed	Junior headcount is unrelated to spending on agents or rises with it
Labor share	Labor share of output in the nonfarm business sector, in percent (BLS Productivity and Costs)	Falls 2 percentage points or more below its 2025 average while labor productivity growth exceeds its 2010–2019 average	Stays within 1 percentage point of its 2025 average while productivity growth exceeds that average
New work	Employment and wage bill in job titles new since 2022, measured as in Autor, Chin, Salomons, and Seegmiller (2024); share of job postings with AI-related titles (Indeed Hiring Lab)	Employment in new titles grows too slowly to offset relative job losses in exposed occupations	Employment and wage bill in new titles grow enough to offset those losses

Source: Author's specification. *Note:* An indicator whose value falls between its two thresholds is classified as inconclusive. The indicators are not independent: the first two share a data source, and the last two respond to the same aggregate shocks.

The clearest single piece of evidence for demand expansion would be joint growth of output and employment in legal services, software publishing, and call centers, the industries whose core tasks AI performs best. The clearest evidence for substitution would be the reverse: output growing in those industries while employment falls, with entry-level hiring leading the decline.

Concluding Remarks

In a standard derived-demand framework, whether cheaper machine intelligence raises or lowers employment depends on whether the elasticity of product demand exceeds the elasticity of substitution between AI and labor—the inequality Hicks identified nearly a century ago. Past automation produced both outcomes. Partial automation in growing markets raised incumbent employment, whole-function substitution lowered it whatever happened to demand, and occupations moved from the first case to the second as a technology's scope broadened.

Read through this condition, three features of the current debate look different. To the extent that the boom in AI use reflects falling effective prices, it is evidence of substitutability, not of job creation, and it raises the bar that the demand-expansion view must clear. The absence of aggregate effects is what the condition predicts while AI's cost share remains small, whichever view is right; it rules out rapid disruption but neither long-run outcome. And the finding that novices gain most from AI copilots does not imply that firms will hire more of them: copilots raise junior employment only if juniors substitute readily for other inputs, and the capability that helps a novice can also replace one.

The two views therefore differ chiefly in two forecasts—of σ, and of whether new tasks for people appear as fast as AI absorbs existing ones—and both depend on the path of AI capabilities. Neither elasticity is beyond measurement: both can be read from the responses of output and labor input to the falling effective price of AI services, once that price and AI's share of costs are measured. The indicators specified here should reveal the direction of change by the end of 2028. The most consequential may be entry-level hiring. Experienced workers are produced from entrants, and a persistent decline in junior hiring would reduce the future supply of the very workers whose complementarity with AI the demand-expansion view relies on.

I thank Geoffrey Hinton for correspondence on the argument of this article. I am employed by Google, which develops and sells artificial intelligence products and services. The views expressed are my own, and no party had the right to review this article before publication.

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Online Appendix

Will Cheaper Machine Intelligence Raise Employment? Demand Expansion, Substitution, and the Early Evidence

Online Appendix A: Derivations

The results in the main text are standard (Hicks 1932; Allen 1938; Hamermesh 1986, 1993). They are derived here in the article's notation, together with the extensions used in the sections on measurement and on junior workers.

Setup. Let C(ω_L, ω_M) be the unit cost function of a constant-returns technology, defined over the effective input prices ω_L = w/a_L and $\omega_M = p_M/a_M$, with partial derivatives C_1, C_2, C_{11}, C_{12}, and C_{22}. By Shephard's lemma, the input of effective labor per unit of output is C_1, so employment is

$$L = \frac{Q\, C_1(\omega_L, \omega_M)}{a_L}. \tag{A1}$$

Competitive pricing gives p = C(ω_L, ω_M), and product demand gives ∂ln Q/∂ln p = $-\varepsilon$. The cost shares are $s_L = \omega_L{\cdot}C_1$/C and $s_M = \omega_M{\cdot}C_2$/C, and the Allen–Uzawa elasticity of substitution is σ = C$\cdot C_{12}/(C_1{\cdot}C_2)$. Because C is homogeneous of degree one, C_1 and C_2 are homogeneous of degree zero, so $\omega_L{\cdot}C_{11} + \omega_M{\cdot}C_{12} = 0$ and $\omega_L{\cdot}C_{12} + \omega_M{\cdot}C_{22} = 0$.

The price of AI services. Holding w and a_L fixed,

$$\frac{\partial \ln L}{\partial \ln \omega_M} = -\varepsilon\, \frac{\omega_M C_2}{C} + \frac{\omega_M C_{12}}{C_1} = s_M(\sigma - \varepsilon). \tag{A2}$$

The second term uses $\omega_M{\cdot}C_{12}/C_1 = \sigma{\cdot}\omega_M{\cdot}C_2$/C = $\sigma{\cdot}s_M$.

Labor-augmenting technical change. Holding w and ω_M fixed, ∂ln ω_L/∂ln $a_L = -1$, so ∂ln p/∂ln a_L = $-s_L$ and

$$\frac{\partial \ln L}{\partial \ln a_L} = \varepsilon\, s_L - \frac{\omega_L C_{11}}{C_1} - 1 = s_L(\varepsilon - 1) + s_M(\sigma - 1). \tag{A3}$$

The second equality uses $-\omega_L{\cdot}C_{11}/C_1 = \omega_M{\cdot}C_{12}/C_1 = \sigma{\cdot}s_M$ and $s_L + s_M = 1$. With labor as the only input ($s_L = 1$), $L = Q/a_L$ and $L \propto a_L^{\varepsilon-1}$.

Demand for AI services and the Jevons condition. Physical AI services are M = Q$\cdot C_2/a_M$. Using $\omega_M{\cdot}C_{22}/C_2 = -\sigma{\cdot}s_L$, the elasticity of effective AI services with respect to their effective price is $-(s_L{\cdot}\sigma + s_M{\cdot}\varepsilon)$, and

$$\frac{\partial \ln M}{\partial \ln a_M} = s_L\,\sigma + s_M\,\varepsilon - 1. \tag{A4}$$

Total use of AI services rises with their efficiency—Jevons's case—if and only if $s_L{\cdot}\sigma + s_M{\cdot}\varepsilon > 1$. The same condition determines whether spending on AI services, $p_M{\cdot}$M, rises when p_M falls, because $\partial\ln(p_M{\cdot}\mathrm{M})/\partial\ln p_M = 1 - (s_L{\cdot}\sigma + s_M{\cdot}\varepsilon)$. When s_M is small, the condition is approximately $\sigma > 1$.

Relation to Marshall's third law. The own-price elasticity of labor demand is $-[(1 - s_L)\sigma + s_L{\cdot}\varepsilon]$. Its magnitude rises with labor's cost share if and only if $\varepsilon > \sigma$. This is Hicks's (1932) qualification of Marshall's (1890) rule that demand for an input is less elastic when the input accounts for a small share of cost. The same inequality determines the sign of (A2).

Decomposition into scale and substitution effects. Write ln L = ln Q + ln(L/Q). From pricing and product demand, $\partial\ln$ Q/$\partial\ln\,\omega_M = -\varepsilon{\cdot}s_M$, and from (A1), $\partial\ln(\mathrm{L/Q})/\partial\ln\,\omega_M = \omega_M{\cdot}C_{12}/C_1 = \sigma{\cdot}s_M$. Defining Δln p as the change in log unit cost (to first order, $s_M{\cdot}\Delta\ln\,\omega_M$),

$$\Delta \ln L = \underbrace{-\,\varepsilon\,\Delta \ln p}_{\Delta \ln Q} + \underbrace{\sigma\,\Delta \ln p}_{\Delta \ln(L/Q)}, \qquad \varepsilon = -\frac{\Delta \ln Q}{\Delta \ln p}, \quad \sigma = \frac{\Delta \ln(L/Q)}{\Delta \ln p}. \tag{A5}$$

For the CES technology below, ln(L/Q) = constant + $\sigma{\cdot}$ln C − $\sigma{\cdot}$ln ω_L − ln a_L, so with w and a_L fixed, (A5) holds exactly for discrete changes when Δln p is the exact change in unit cost. If wages or labor productivity also change, σ is the ratio of $\Delta\ln(a_L{\cdot}\mathrm{L/Q})$ to $\Delta\ln(\mathrm{C}/\omega_L)$. If pass-through to the output price is incomplete, ε should be computed against the observed change in the output price and σ against the change in unit cost.

Factor shares. For the CES technology $Q = [\alpha(a_L L)^\rho + (1-\alpha)(a_M M)^\rho]^{1/\rho}$, with $\sigma = 1/(1-\rho)$, cost minimization gives

$$\frac{s_M}{s_L} = \left(\frac{1-\alpha}{\alpha}\right)^{\sigma} \left(\frac{\omega_M}{\omega_L}\right)^{1-\sigma}, \qquad \Delta \ln \frac{s_M}{s_L} = (1-\sigma)\,\Delta \ln \frac{\omega_M}{\omega_L}. \tag{A6}$$

If $\sigma > 1$, a fall in ω_M raises AI's share of cost and lowers labor's. The second expression is the estimating equation for the cost-share route in Table 2 of the article.

Large price changes. With CES technology, $s_L = \alpha^\sigma(\omega_L/C)^{1-\sigma}$, so with ω_L fixed, $\Delta \ln s_L = (\sigma - 1)\,\Delta \ln C$. Substituting into (A5), with Δln p = Δln C, gives

$$\Delta \ln L = \frac{\sigma - \varepsilon}{\sigma - 1}\,\Delta \ln s_L. \tag{A7}$$

Because s_L is close to one when AI's share is small, $\Delta\ln\,s_L$ is approximately the negative of the change in AI's share, so the employment effect is of the same order unless σ is close to one or $|\sigma - \varepsilon|$ is large relative to $|\sigma - 1|$. Figure 2 of the article plots $\exp\{[(\sigma-\varepsilon)/(\sigma-1)]\ln(1-s_M)\} - 1$,

the change in employment as AI's share rises from zero to s_M. When $\sigma = 1$, AI's share is constant and (A2) integrates directly to $\Delta\ln \mathrm{L} = s_M(1-\varepsilon)\cdot\Delta\ln\,\omega_M$, which can be large for a large price decline.

Several types of labor. Let the inputs be junior labor J, other labor, and AI services M, with cost shares s_k and Allen partial elasticities of substitution $\sigma_{jk} = \mathrm{C}\cdot C_{jk}/(C_j\cdot C_k)$. The elasticity of the conditional demand for input j with respect to the effective price of input k is $s_k\cdot\sigma_{jk}$, and homogeneity implies $s_J\cdot\sigma_{JJ} = -\Sigma_{k\neq J}\, s_k\cdot\sigma_{Jk}$. Repeating the steps behind (A2) and (A3) gives

$$\frac{\partial \ln L_J}{\partial \ln \omega_M} = s_M(\sigma_{JM} - \varepsilon), \tag{A8}$$

$$\frac{\partial \ln L_J}{\partial \ln a_J} = s_J(\varepsilon - 1) + \sum_{k\neq J} s_k(\sigma_{Jk} - 1). \tag{A9}$$

The expression in the article's section on junior workers writes the sum in (A9) as $(1-s_J)(\sigma_J - 1)$, where σ_J is the cost-share-weighted average of the σ_{Jk}. Equation (A8) governs the agent channel and (A9) the copilot channel; the experiments discussed in the article estimate the productivity change that enters (A9), not σ_{JM}.

A task-based interpretation. Suppose that output combines a continuum of tasks with a cross-task elasticity of substitution η, and that each task is performed by labor or by AI, whichever is cheaper in that task. At a fixed assignment of tasks, the elasticity of substitution between labor and AI equals η. When AI's effective price falls, tasks at the margin switch from labor to AI, so the reduced-form σ in (A2) exceeds η by an amount that depends on how many tasks lie near the cost threshold (Acemoglu and Restrepo 2018). An advance in capabilities that makes new tasks feasible for AI at any price is not a movement along this schedule; in the reduced form it appears as a fall in labor's task content, Γ, in (A10) below.

New tasks and the task content of production. Following Acemoglu and Restrepo (2019), write the CES unit cost as $C^{1-\sigma} = \Gamma\,\omega_L^{1-\sigma} + (1-\Gamma)\,\omega_M^{1-\sigma}$, where Γ is labor's task content of production. This is the technology behind (A6), with its distribution weights rescaled to sum to one. Shephard's lemma gives $C_1 = \Gamma\,\omega_L^{-\sigma}C^{\sigma}$, so by (A1), $\ln L = \ln Q + \ln\Gamma - \sigma\ln\omega_L + \sigma\ln C - \ln a_L$. With w and a_L fixed and $Q \propto C^{-\varepsilon}$,

$$\Delta\ln L = (\sigma - \varepsilon)\,\Delta\ln C + \Delta\ln\Gamma, \tag{A10}$$

which holds exactly for discrete changes. With Γ fixed, (A10) reduces to (A5). A change in Γ also changes unit cost: automating a task that AI performs more cheaply lowers both Γ and C, so its effect on employment combines displacement ($\Delta\ln\Gamma < 0$) with a cost effect, $(\sigma-\varepsilon)\cdot\Delta\ln$ C, which is positive when $\varepsilon > \sigma$. New tasks in which labor has the advantage raise Γ, the reinstatement effect. Γ summarizes the structure of production; it is not an elasticity, and its path depends on the direction of innovation rather than on prices.

Perfect substitution in a task. If one unit of labor performs a_L units of a task and one unit of AI services performs a_M units, cost minimization assigns the task to labor only if $w/a_L \leq p_M/a_M$, that is, $w \leq (a_L/a_M) \cdot p_M$. Near this threshold, the cost saving from automating the task is small, so the price reduction that drives the scale effect in (A5) is small while the displacement of labor in the task is complete.

Scope. All results are for one competitive industry with constant returns and an exogenous wage. Apart from (A10), they abstract from new tasks; all of them abstract from reallocation across industries, and from changes in σ itself, which the main text discusses.

Online Appendix B: Coding of Table 1 and Two Case Studies

Table 1 of the article codes each episode on three dimensions, using the sources listed in the table.

Scope automated. An episode is coded as automating the entire function, or the core function, when the technology produced the output that defined the occupation without a member of the occupation operating it: traction for horses and mules, calculation for human computers, recording a likeness for miniature portrait painters, connecting calls for telephone operators, operating the car for elevator operators, and recording and reconciling transactions for bookkeeping clerks. It is coded as automating a subset of tasks when members of the occupation continued to perform other tasks that made up a substantial part of the job, as accountants did after the spreadsheet and tellers did after the automated teller machine. Online and mobile banking is grouped with whole-function substitution because it performs most routine transactions without a teller.

Demand for output. "Expanded" means that output of the good or service grew faster than the productivity gain the technology delivered, so that total input into the activity rose. "Grew less than productivity" means that output grew but total input fell, as in agriculture after mechanization. "No induced demand" means that the technology did not make the final service cheap enough to raise its consumption appreciably, as with elevator rides. The judgments are qualitative; no elasticities were estimated.

Incumbent employment. The change in employment of the incumbent occupation over the period shown, from the cited source.

Placement in Figure 1. Episodes coded as automating the entire or core function are placed in the row for whole-function substitution, and those coded as automating a subset of tasks in the row for partial automation. Episodes coded "Expanded" are placed in the column for demand growth, as is online and mobile banking, which the author codes as demand met through new channels rather than by tellers; farm mechanization and automatic elevators are placed in the other column. Tellers appear twice, under partial automation for 1988–2004 and under whole-function substitution after 2010, to illustrate the migration of an occupation between cells. Radiology and telephone switching are not placed, for the reasons given in the note to Figure 1.

The two case studies below expand on episodes in Table 1. Software development and photography both produced large expansions in demand, but they differ in which occupation performed the new

work, which makes them a useful pair.

Software development: the incumbent occupation gained. Programming productivity has risen by large multiples. The first FORTRAN compiler, released in 1957, reduced the number of statements needed to program a machine by a factor of roughly 4 to 20; in one benchmark, 47 FORTRAN statements produced about 1,000 machine instructions (Backus et al. 1957). Compilers, open-source libraries, and cloud services have added further multiples since. Employment grew nevertheless. US employment in information technology occupations rose from 450,000 in 1970 to 4.6 million in 2014 (Beckhusen 2016), and software developers held about 1.7 million jobs in 2025 (BLS 2026c). Because employment equals output divided by output per worker, a tenfold rise in employment alongside large productivity gains implies that output grew by a far larger factor than output per worker.

Earnings rose with employment. Among men working full time, year-round, real median earnings in nonmanagerial information technology occupations rose from $74,180 in 1970 to $80,895 in 2014, in 2014 dollars, while median earnings for all such men fell from $55,690 to $49,150 (Beckhusen 2016). The content of the work has shifted toward design. The Bureau of Labor Statistics projects that employment of computer programmers will fall about 7 percent between 2025 and 2035, while employment of software developers will rise about 10 percent (BLS 2026a, 2026c). Software development is the clearest historical case of complementary expansion, and it is now among the occupations most exposed to AI.

Photography: a new occupation gained. Photography did not raise the productivity of miniature portrait painters; it replaced them in recording a likeness. In the 1840s, a daguerreotype portrait cost $2 to $5 in most American towns (Nelson 1996), within reach of buyers who could not afford a painted miniature, and after the invention of the daguerreotype in 1839 many American miniaturists abandoned their art (Barratt 2004).

Demand for photographic portraits then grew rapidly. By one estimate, 300 to 400 million cartes de visite were sold each year in England in the early-to-mid 1860s (Darrah 1981, cited in Moran 2018), when cartes typically sold for $2 to $3 a dozen (Darrah 1981). Census returns record 51 professional photographers in Britain in 1851 (Shrimpton n.d.) and 2,366 people employed in photography in England and Wales in 1861 (Edwards 2006). The technology produced complementary expansion for image-making as an activity and whole-function substitution for the incumbent occupation. Unlike AI, however, it still required a human operator for every portrait.

Online Appendix C: Studies of AI and Labor Markets since 2022

Table C1 summarizes, in a common format, the studies of AI and labor markets cited in the article, together with several related studies not discussed there. Panels are ordered by the strength of the research design for causal inference: experiments first, then comparisons of more and less exposed units with a defined control group, then descriptive comparisons, then surveys and usage data. A final panel lists studies of earlier labor-saving technologies cited for comparison. The last column

states which part of the derived-demand framework each study bears on. Most of the studies are working papers or policy reports and may change; Imas and Schaal (2026) maintain a periodically revised review of this literature.

Table C1

Studies of AI and Labor Markets, by Research Design

Study	Design	Data	Main finding	Bears on
Panel A. Experiments				
Brynjolfsson, Li, and Raymond (2025)	Staggered rollout of an AI assistant	Customer-support agents at one firm	Productivity up about 15 percent on average and about 30 percent for less experienced and less skilled agents; minimal gains for the most skilled; suggestive evidence that the assistant disseminated top agents' practices	Copilot channel, by experience
Noy and Zhang (2023)	Randomized experiment	453 college-educated professionals, writing tasks	Time per task down 40 percent; quality up 18 percent; gap between stronger and weaker writers narrowed	Copilot channel
Dell'Acqua et al. (2023, 2026)	Randomized experiment	758 consultants	Within the model's capabilities, performance up 43 percent for below-average and 17 percent for above-average consultants (2023 version); outside them, correct answers 19 percentage points less likely	Variation of σ across tasks
Cui et al. (2026)	Three randomized field experiments	Nearly 5,000 software developers at three companies	Completed tasks up about 26 percent; larger point estimates for junior developers, not significantly different	Copilot channel, by experience
Peng, Kalliamvakou, Cihon, and Demirer (2023)	Randomized experiment	Programmers given a coding task, with or without a coding assistant	Treated programmers completed the task 55.8 percent faster	Copilot channel
Agarwal, Moehring, Rajpurkar, and Salz (2023)	Randomized experiment	Radiologists given AI predictions	No improvement in average performance; best design assigned each case to the AI or to the radiologist alone	Case-by-case substitution
Panel B. Comparisons with a control group				

Study	Design	Data	Main finding	Bears on
Hui, Reshef, and Zhou (2024)	Difference-in-differences	Large online freelancing platform	In exposed occupations, monthly jobs down about 2 percent and monthly earnings about 5 percent; higher-quality freelancers not protected	σ in a market with few frictions
Demirci, Hannane, and Zhu (2025)	Difference-in-differences	Freelancing platform job posts	Writing and coding posts down 21 percent relative to manual-intensive jobs within eight months	σ in a market with few frictions
Humlum and Vestergaard (2026)	Difference-in-differences	Survey of about 25,000 Danish workers in 11 exposed occupations, linked to administrative records	Effects on earnings or hours larger than 2 percent ruled out two years after ChatGPT; no differential change in employment or hiring composition at adopting workplaces	Aggregate net effect
Nunn (2026)	Synthetic difference-in-differences	US monthly employment microdata	Effects on employment shares and real wages of exposed occupations not distinguishable from zero	Aggregate net effect
Hosseini Maasoum and Lichtinger (2025)	Adopters versus nonadopters	Résumé data, about 65 million workers at more than 280,000 US firms	Junior employment at adopters down roughly 8–10 percent within six to eight quarters, mainly through slower hiring; senior employment rising	Hiring margin; σ by experience
Kharazian, Simon, and Stevens (2026)	Event study, early versus later adopters	AI spending of 21,559 US firms linked to Revelio Labs headcount	High-intensity adopters: headcount up 10.2 percent and entry-level headcount up 12.0 percent within two years; an association, with some pre-trends	Firm-level $\varepsilon - \sigma$, including business stealing
Hampole, Papanikolaou, Schmidt, and Seegmiller (2025)	Exposure within and across firms	Firm and occupation data	Exposure lowers demand for exposed tasks; productivity-driven expansion offsets much of the decline	Both terms of the condition, within firms
Brynjolfsson, Chandar, and Chen (2025, 2026)	Exposure comparisons; firm-by-time controls	ADP payroll records; exposure of Eloundou, Manning, Mishkin, and Rock (2024)	Ages 22–25 in the two most exposed quintiles 19 percent below the trend of the three least exposed as of June 2026; with firm-by-time controls, significant from 2024 and not explained by interest-rate sensitivity; remote-work control changes the estimate little (-0.179 to -0.152); declines concentrated in occupations with high Anthropic Economic Index automation shares, not augmentation shares	Hiring margin; attribution; copilot versus agent channel

Study	Design	Data	Main finding	Bears on
Tucker (2026)	Exposure comparisons across industries	Quarterly Workforce Indicators, more than 95 percent of private-sector jobs	Early-career employment in the most exposed industries down about 12 percent in relative terms over ten quarters; hiring largely recovered by early 2025; monetary policy explains at most a quarter	Hiring margin
Orr, Tucker, and Warren (2026)	Exposure comparisons across college majors	Census-linked records for 2016–2024 bachelor's graduates	Most exposed decile of majors: initial employment 5 percentage points lower, initial earnings 13 percent lower	Hiring margin
Lambert and Schindler (2026)	Exposure comparisons with remote-work controls	243 million new hires (Revelio Labs) and 407 million job postings (Lightcast), four countries, 2017–2025	Controlling for remote-work exposure sharply attenuates the AI coefficient, often to statistical insignificance	Attribution
Frank et al. (2026)	Timing of deterioration by exposure	US unemployment insurance records, LinkedIn profiles, course syllabi	Unemployment risk in exposed occupations rose from early 2022; entry fell from the 2021 graduating cohort	Attribution
Facius and Iacono (2026)	Exposure comparisons	Norwegian register data, 2015 to March 2025	No robust displacement of young workers; declines partly reflect earlier trends	Hiring margin
Hernæs and Kostøl (2026)	Exposure comparisons	Norwegian private-sector employment, 2021 to February 2026	No overall gap; emerging gap for ages 22–25 from spring 2025; software developers aged 21–30 down about 18 percent per capita (descriptive)	Hiring margin
Kauhanen and Rouvinen (2026)	Exposure comparisons	Finnish Incomes Register, 2019–2025	No displacement of young workers linked to AI exposure	Hiring margin
Fairlie and Wu (2026)	Exposure comparisons	Current Population Survey, summers 2022–2026	Unemployment of 22- to 25-year-old graduates 7.3 percent in summer 2026, within the 2022–2025 range of 6.3–7.8 percent; no rise relative to older graduates	Hiring margin
Eisfeldt, Schubert, and Zhang (2023)	Event study of stock returns	US listed firms	Firms with more exposed workforces earned higher returns after ChatGPT's release	Expected labor savings or productivity gains
Acemoglu, Autor, Hazell, and Restrepo (2022)	Exposure comparisons, earlier AI	US online job vacancies, 2010s	Exposed establishments reduced non-AI hiring; effects on occupations and industries too small to detect	Precedent for aggregate nulls

Study	Design	Data	Main finding	Bears on
Panel C. Descriptive comparisons				
Gimbel, Kinder, Kendall, and Lee (2025)	Tracking of occupational mix	US labor market data	No discernible disruption in the 33 months after ChatGPT; occupational mix shifting only slightly faster than in earlier periods, beginning before generative AI	Aggregate net effect
Mahoney, McEntarfer, and Wahal (2026)	Unemployment by exposure quintile	Current Population Survey, 2015–2026; exposure of Felten, Raj, and Seamans (2021)	Since 2022, unemployment up 0.77 percentage points in the top exposure quintile and 0.85 points in the bottom quintile; new-graduate unemployment 5.6 percent in early 2026	Aggregate net effect
Massenkoff and McCrory (2026)	Unemployment by observed AI use	US unemployment data	No systematic rise in unemployment for exposed workers; job-finding rate of 22- to 25-year-olds into the most exposed occupations about 14 percent below 2022, barely significant	Aggregate net effect; hiring margin
Atkinson and Yamco (2026)	Flows by exposure	Current Population Survey	Decline confined to ages 20–24 and operating through lower inflows; about 0.1 percentage point of aggregate unemployment	Hiring margin
Gallacher (2026)	Job postings	Indeed	US software-development postings up almost 15 percent since February 2025, against about −7 percent overall; still 27.5 percent below February 2020; senior roles 71 percent of the increase	Composition by experience
Aidala and Puri (2026)	Job postings by seniority	Indeed	All occupations, year to May 2026: senior postings up 14.7 percent, entry-level down 7.5 percent; entry-level roles 4.5 percent of software postings in the first quarter of 2026	Composition by experience
Panel D. Surveys and usage data				
Bick, Blandin, and Deming (2026)	Nationally representative survey	US workers, 2024	1–5 percent of all work hours assisted by generative AI	Scale of AI in production
Federal Reserve Bank of St. Louis (2026)	Quarterly Real-Time Population Survey	US employed adults, 2024–2026	Weekly use of generative AI for work up from 28.2 percent (third quarter of 2024) to 39.2 percent (second quarter of 2026); hours saved up from 1.6 to 2.2 percent of all work hours	Scale of AI in production; growth of use

Study	Design	Data	Main finding	Bears on
Humlum and Vestergaard (2025)	Survey linked to administrative records	Danish workers in exposed occupations	Users' time savings about 2.8 percent of work hours; 3–7 percent of productivity gains passed through to earnings	Scale of AI in production
Grundy, Breaux, and Khatiwoda (2026)	Business Trends and Outlook Survey	US firms	17–20 percent of firms used AI between December 2025 and May 2026	Scale of AI in production
Yotzov et al. (2026)	Executive survey	Nearly 6,000 executives in the United States, United Kingdom, Germany, and Australia	69 percent of firms use AI; more than 90 percent report no employment impact over the past three years; in the United Kingdom, about two-thirds of expected cuts would come through reduced hiring	Aggregate net effect; hiring margin
James and Pendell (2026)	Workforce survey	Gallup, first quarter of 2026	1 percent of laid-off workers named AI or automation as the primary cause	Aggregate net effect
Handa et al. (2025)	Classification of conversations	Anthropic consumer chatbot	57 percent of sampled conversations augmented a user's work and 43 percent automated it	Copilot versus agent channel
Appel et al. (2025)	Classification of usage	Anthropic consumer and programming-interface traffic	Automation overtook augmentation; about three-quarters of programming-interface traffic involved automation	Copilot versus agent channel
Panel E. Earlier labor-saving technologies				
Dauth, Findeisen, Suedekum, and Woessner (2021)	Exposure across local labor markets	German workers, 1994–2014	Robots did not raise incumbents' risk of job loss; manufacturing declines came through fewer young entrants; services fully offset	Hiring margin
Feigenbaum and Gross (2024)	Staggered adoption across cities	US census records, 1920–1940	Incumbent telephone operators harmed; later cohorts absorbed by clerical and service jobs	Hiring margin
Bessen, Goos, Salomons, and van den Berge (2025)	Event study of automation spikes	Dutch firm and worker records	Incumbents at automating firms lose a cumulative 9 percent of one year's earnings over five years	Incumbent losses

Source: Author's compilation from the studies listed. *Note:* "Exposure comparisons" contrast more and less AI-exposed occupations, industries, or majors over time. Designs in Panel B differ in how convincingly they rule out other shocks; the article's section on attribution discusses the main threats.

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